

An unexpected role for FosB in activation-induced cell death of T cells

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The CD95 (APO-1/Fas) system plays a major role in induction of apoptosis in lymphoid and nonlymphoid tissues. The CD95 (APO-1/Fas) ligand (CD95L) is induced in response to a variety of signals including TCR/CD3 stimulation or application of chemotherapeutic drugs. Here we report that an AP-1 site located in the 5' untranslated region of the CD95L gene is required for TCR/CD3-mediated induction of the human CD95L promoter. Electrophoretic mobility shift assays using nuclear extracts of Jurkat T cells as well as TCR/CD3-restimulated primary human T cells demonstrated specific binding of AP-1, predominantly composed of c-Jun and FosB, to this sequence. Ectopic expression of transdominant negative Jun mutants strongly reduced CD95L promoter activity and activation-induced cell death (AICD), confirming the functional significance of FosB/c-Jun binding. Thus, our results demonstrate an important novel function for FosB dimerized with c-Jun in TCR/CD3-mediated AICD in human T cells.

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Death receptors, members of the TNFR/nerve growth factor receptor superfamily, play a central role in induction of apoptosis. Among them, CD95 (APO-1/Fas) exerts a major function in induction of apoptosis in activated T cells upon TCR/CD3 restimulation, resulting in activation-induced cell death (AICD). This process represents the downphase of an immune response, which is necessary to maintain immune homeostasis and self-tolerance in the immune system (Dhein *et al.*, 1995). TCR/CD3 restimulation leads to the induction of CD95 ligand (CD95L) expression. Apoptosis is induced in CD95-bearing cells via the formation of a death-inducing signaling complex (DISC) and initiation of a signaling cascade of caspases finally leading to cell death (Schmitz *et al.*, 2000). It has

been shown previously that drug treatment can also induce apoptosis via the CD95/CD95L system (Friesen *et al.*, 1996; Eichhorst *et al.*, 2000).

Upon TCR/CD3 stimulation, the binding of several transcription factors like NF-AT (Holtz-Heppelmann *et al.*, 1998; Li-Weber *et al.*, 1998, 2000; Norian *et al.*, 1998; Kasibhatla *et al.*, 1999), Egr (early growth response protein) (Mittelstadt and Ashwell, 1998; Li-Weber *et al.*, 1999; Mittelstadt and Ashwell, 1999) and IRF-1 (Chow *et al.*, 2000) at the CD95L promoter in Jurkat T cells has been described. Several lines of evidence strongly suggest that, in addition to these factors, members of the Fos, Jun and ATF proteins, which form the transcription factor AP-1, are critically involved in CD95L expression and apoptosis. For example, upon TCR/CD3 stimulation, JNK/SAPKs become activated to hyperphosphorylate and thereby activate c-Jun (Su *et al.*, 1994). Furthermore, p38 MAPK and JNK were found to be involved in CD95L expression upon TCR/CD3 triggering in the murine system (Faris *et al.*, 1998; Zhang *et al.*, 2000). In line with these data, MEKK and JNK/SAPK kinases mediating hyperphosphorylation, and thereby activation of the protooncogene c-Jun, participate in AP-1-dependent activation of CD95L gene expression upon genotoxic stress in murine and human cells (Kasibhatla *et al.*, 1998; Eichhorst *et al.*, 2000; Kolbus *et al.*, 2000; Chang and Karin, 2001). While JunB and JunD have been linked to pro- and antiapoptotic activities, respectively (Passegue and Wagner, 2000; Weitzman *et al.*, 2000), both activities have been found for c-Fos *in vivo* and *in vitro* in various cell types (Hafezi *et al.*, 1997; Ivanov and Nikolic-Zugic, 1997). There is evidence that c-Fos and c-Jun represent the main regulators of CD95L expression in response to genotoxic agents (Kasibhatla *et al.*, 1998; Eichhorst *et al.*, 2000). However, the exact composition of the AP-1 dimer involved in AICD in T cells is presently unknown.

In this study, we have identified an AP-1 site in the 5' UTR (untranslated region) of the human CD95L gene that is involved in TCR/CD3-mediated CD95L induction. Strong binding of AP-1 dimeric complexes composed of FosB and c-Jun was detectable in extracts from Jurkat as well as primary activated human T cells upon TCR/CD3 stimulation. Mutation of the AP-1 site in the 5' UTR strongly reduced TCR/CD3-mediated CD95L promoter induction. Expression of dominant

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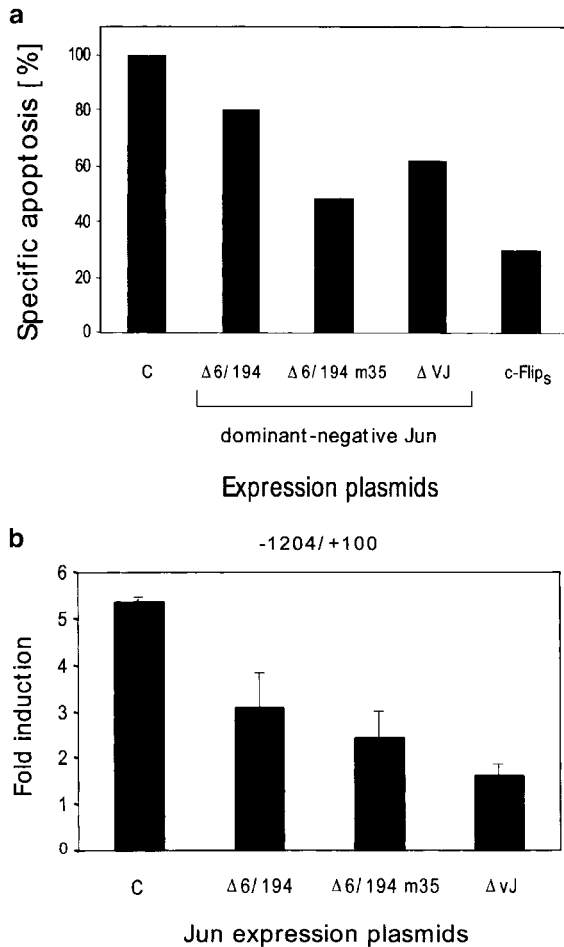


Figure 1 Dominant negative Jun mutants reduce TCR/CD3-mediated CD95L promoter activity. **(a)** The previously described (Li-Weber *et al.*, 1998) human T-lymphoblastoid cell line Jurkat was transiently transfected with 10 μ g of an expression plasmid encoding GFP in the presence of 10 μ g of the indicated expression vectors or empty vector (c). Mutants contain the bZip region of c-Jun ($\Delta 6/194$) or v-Jun (ΔVJ), but lack the transactivation domain (Stein *et al.*, 1993). Mutant $\Delta A6/194m35$ contains additional amino-acid substitutions in the leucine zipper enhancing homo- and heterodimer formation. (H van Dam, S Müller and P Angel, unpublished). GFP-positive cells were sorted 36 h after transfection; 12 h after sorting, living cells were isolated by Ficoll gradient centrifugation, recovered for 7 h, and TCR/CD3-stimulation was performed for a further 36 h. Alternatively, cells were cultivated for 12 h in the presence of LZ-CD95L. Cell death was quantified by FACS analysis (FSC/SSC), gating on GFP-positive cells. **(b)** In all, 10 μ g of a reporter construct containing 1204 bp 5' of the CD95L transcription start and 100 bp 5' UTR in front of firefly luciferase ($-1204/+100$), previously described (Li-Weber *et al.*, 1998; Eichhorst *et al.*, 2000), was transiently transfected into Jurkat T cells in the presence of 5 μ g of empty vector (c) or expression plasmids encoding the indicated dominant negative versions of Jun. Transfection was performed by using a Bio-Rad Gene Pulser (Bio-Rad GmbH, Munich, Germany) set at 960 μ F, 240 V. To compare transfection efficiency, 3 μ g of Renilla luciferase expression vector (Promega, Heidelberg, Germany) was added to the transfection mixture. Transfected cells recovered for 36 h, were stimulated via TCR/CD3 or left untreated for a further 12 h. Then cells were lysed using the lysis buffer contained in the dual luciferase assay system (Promega, Heidelberg, Germany). Luciferase activity was measured as described in the manufacturer's instructions using the DuoLumat LB9507 (Berthold, Wildbach, Germany). Protein determination was performed using the Bio-Rad protein assay (Bio-Rad GmbH, Munich, Germany) to normalize protein content. The results are representative of three independent experiments performed in triplicate.

negative mutants of c-Jun interfering with the function of FosB/c-Jun dimers strongly reduced TCR/CD3-mediated CD95L promoter activity as well as AICD. Our results contribute to the understanding of the complex transcriptional regulation of the human CD95L upon TCR/CD3 and identify a yet unknown function of FosB in AICD.

To test whether AP-1 is involved in AICD of T cells, we transiently transfected Jurkat T cells with expression vectors encoding dominant negative mutants of Jun, c-FLIP_s or empty vector together with a GFP expression plasmid. Apoptosis of GFP-positive cells was determined after 36 h of TCR/CD3 stimulation. Expression of the indicated dominant negative Jun mutants led to a reduction of TCR/CD3-mediated apoptosis (Figure 1a). These results demonstrate that c-Jun or one of its dimerization partners are critically involved in TCR/CD3-induced apoptosis.

Recently, we reported that AP-1 is involved in CD95L induction upon drug treatment in human liver cells, resulting in apoptosis (Eichhorst *et al.*, 2000). Since AICD is blocked in the presence of dominant negative Jun, we tested whether this is due to reduced TCR/CD3-mediated CD95L expression. Jurkat T cells were transiently transfected with a reporter construct containing a 1204 bp fragment 5' of the transcription start and 100 bp of the 5' UTR ($-1204/+100$) of the human CD95L gene fused to luciferase. This reporter construct was induced five fold upon TCR/CD3 stimulation. Coexpression of dominant negative Jun mutants reduced CD95L induction compared to transfections with empty vector (Figure 1b). These results demonstrate that c-Jun or one of its interacting proteins are involved in TCR/CD3-induced CD95L promoter activity.

Drug treatment induces binding of AP-1 (c-Jun/c-Fos) to a novel AP-1 site in the 5' UTR of the CD95L gene, which is necessary for CD95L induction (Eichhorst *et al.*, 2000). Therefore, we asked whether this site may also be responsible for TCR/CD3-mediated CD95L induction. EMSAs using the sequence containing this AP-1 recognition sequence of the 5' UTR ($+75/+96$) resulted in strong induction of a protein/DNA complex within 8 h of TCR/CD3 stimulation in nuclear extracts of Jurkat (Figure 2a, lane 4) and restimulated day 6 primary human T cells (Figure 2a, lane 6). Competition with 50-fold molar excess of an oligonucleotide containing the wild-type AP-1 site prevented the formation of this complex (data not shown). Addition of c-Jun-specific antibodies resulted in a supershifted complex (Figure 2b, lanes 3 and 7). In the presence of an antiserum recognizing all members of the Fos family, the induced complex disappeared in EMSAs with extracts from Jurkat as well as primary restimulated T cells (Figure 2c, lane 12, Figure 2d, lane 3). Surprisingly, the presence of c-Fos-specific antibodies did not influence the complex (Figure 2c, lane 10, Figure 2d, lane 7) but efficiently affected the electrophoretic mobility of *in vitro* translated c-Jun and c-Fos (data not shown). When we included antibodies recognizing individual members of the Fos protein family, only the addition of FosB-specific antibodies inhibited the

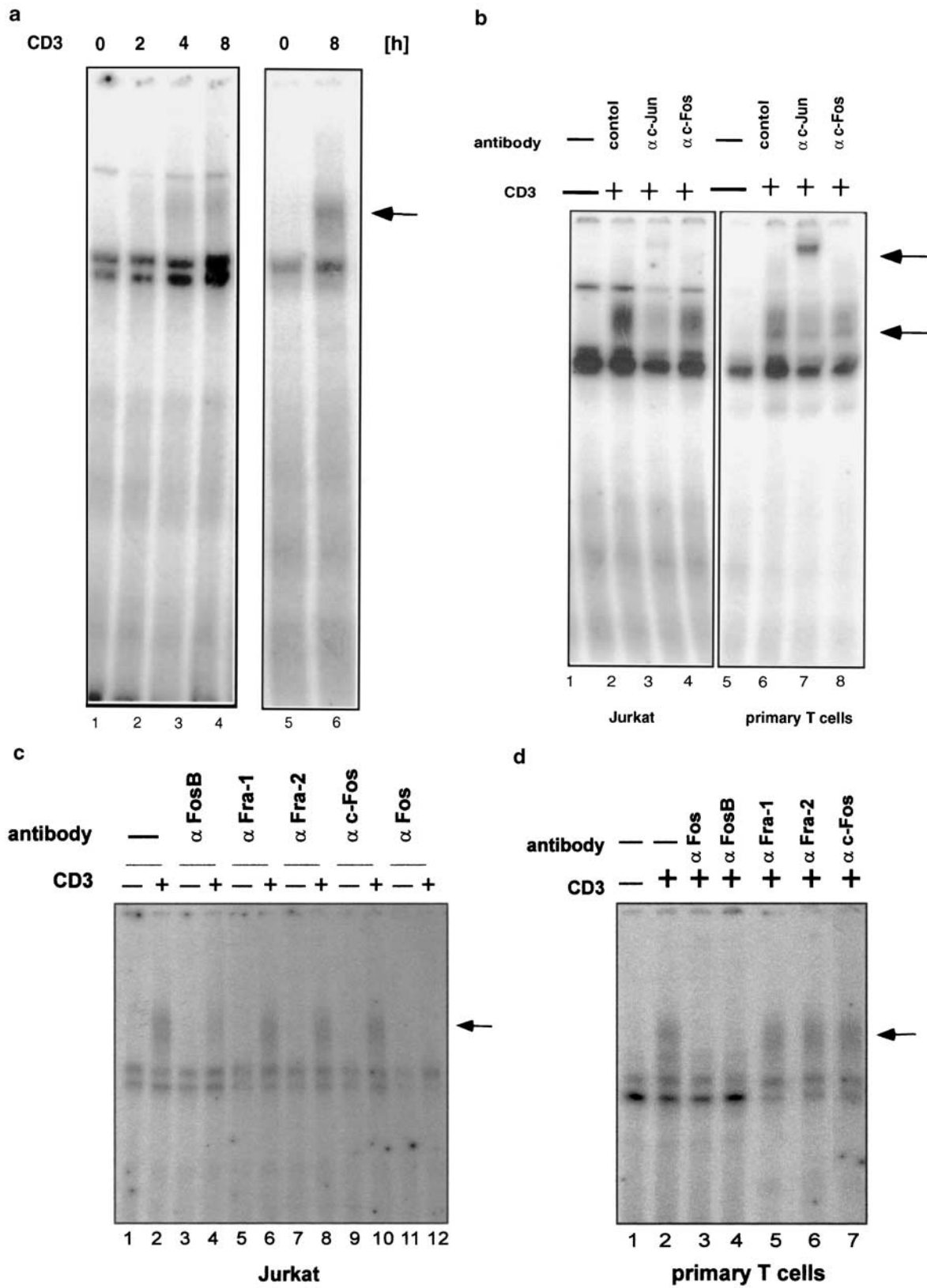


Figure 2 (Continued)

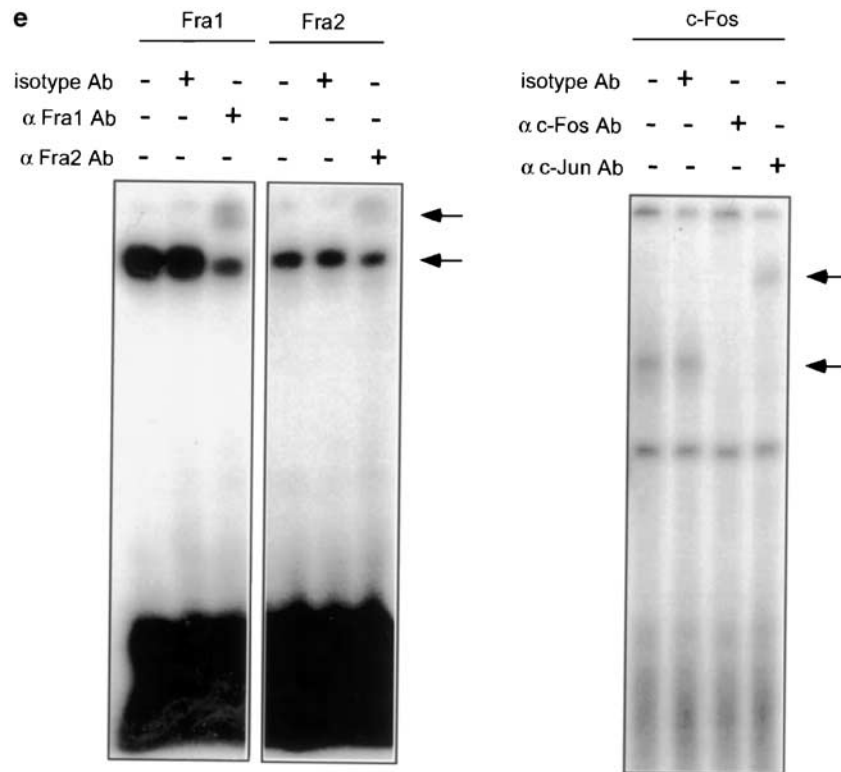


Figure 2 AP-1 binding in the 5' UTR is induced upon TCR/CD3 stimulation. (a) Human peripheral T cells were prepared and activated as described (Kirchhoff *et al.*, 2000). EMSAs were performed using nuclear extracts of Jurkat T cells stimulated via TCR/CD3 (CD3) or 6 days activated primary T cells restimulated via TCR/CD3 for the indicated time periods using plate bound anti-CD3 mAbs OKT3 (30 μ g/ml) and subsequent crosslinking with 2 μ g/ml goat anti-mouse IgG (Pharmingen, Hamburg, Germany). Nuclear extracts were prepared according to the method of Andrews and Fallier (1991). Protein content was determined using the Bio-Rad protein assay. *In vitro* translated proteins were generated using the TNT Quick Coupled Transcription/Translation Systems (Promega, Heidelberg, Germany). In all, 2.5 pmol double-stranded oligonucleotides comprising the AP-1 site at +90 in the CD95L promoter were end-labeled with T4 polynucleotide kinase (Roche, Germany) using 25 000 Ci/mmol [γ - 32 P]ATP (Amersham GmbH, Braunschweig, Germany). Sequences of the single-stranded oligonucleotides were +76/+95 sense (5' GAT CGGCTGGCCTGACTCACCAGCA 3') and +76/+95 antisense (5' GATCTGCTGGT-GAGTCAGGCCAGCCG 3'). Binding reactions were carried out at 4°C for 30 min using 2.5–5 μ g nuclear proteins or 0.5 μ l of *in vitro* translated proteins in a buffer containing 10 mM HEPES, pH 7.5, 0.5 mM EDTA, 100 mM KCl, 2 mM DTT, 2% glycerol, 4% Ficoll 400, 0.25 % NP-40 and 0.1 μ g/ μ l poly dI/dC in a total of 30 μ l containing 50 fmol of labeled oligonucleotides. For supershift analysis, 1 μ g of antibody was added to the binding reaction. Samples were analysed on a 6% nondenaturing PAGE in 1% Tris-borate-EDTA. (b) As described in (a) using nuclear extracts of TCR/CD3-stimulated Jurkat T cells or restimulated day 6 activated primary T cells (8 h) in the presence of abs specific for c-Jun or c-Fos or an isotype control (control). (c) EMSAs using nuclear extracts of TCR/CD3 stimulated Jurkat T cells or (d) restimulated day 6 activated primary T cells (8 h) in the presence of abs specific for all Fos family members. α Fos ab: directed against all Fos family members. The antibodies c-Jun (H-79), c-Fos (SC-52), FosB (H-75), Fra1 (R-20) and Fra2 (Q-20) were purchased from Santa Cruz Biotechnology Inc. (Heidelberg, Germany). (e) EMSAs using *in vitro* translated Fra1, Fra2 or c-Fos in combination with c-Jun in the presence or absence of abs specific for Fra1, Fra2, c-Fos, c-Jun or an isotype control

induced complex in EMSAs with extracts from Jurkat (Figure 2c, lane 4) as well as primary T cells (Figure 2d, lane 4). Controls for the antibodies recognizing c-Jun or Fos-family members used in Figures 2b–d are shown in Figure 2e. These results demonstrate that TCR/CD3 (re)stimulation induces AP-1 binding, composed of c-Jun and FosB, to the 5' UTR of the CD95L gene.

To confirm the absolute requirement of the AP-1 site identified above in TCR/CD3-mediated CD95L gene induction by a functional assay, we compared induction levels of the –1204/+100 and –36/+100 reporter construct containing either wild-type or the mutated AP-1 site in Jurkat T cells. As indicated in Figure 3a, TCR/CD3-mediated induction of the wild-type –1204/+100 construct (sixfold) was significantly reduced upon mutation of the AP-1 site (2.5-fold). Similarly, the two

fold induction of the –36/+100 minimal promoter was completely abolished in its mutant counterpart. However, although FosB is essential for proper induction of the CD95L promoter, FosB alone is not capable of activating the promoter in an overexpression system (Figure 3b), suggesting that FosB is present in the T cell all the time and c-Jun, as the dimerization partner is the limiting step. These results demonstrate that the AP-1 site in the 5' UTR of the human CD95L gene plays an important role in TCR/CD3-mediated induction of CD95L.

In the present study, we demonstrate for the first time that AP-1, composed of FosB and c-Jun, is necessary for AICD of human T cells. We identified TCR/CD3-induced AP-1 binding to a sequence in the 5' UTR of the human CD95L promoter. Mutation of this FosB/c-Jun

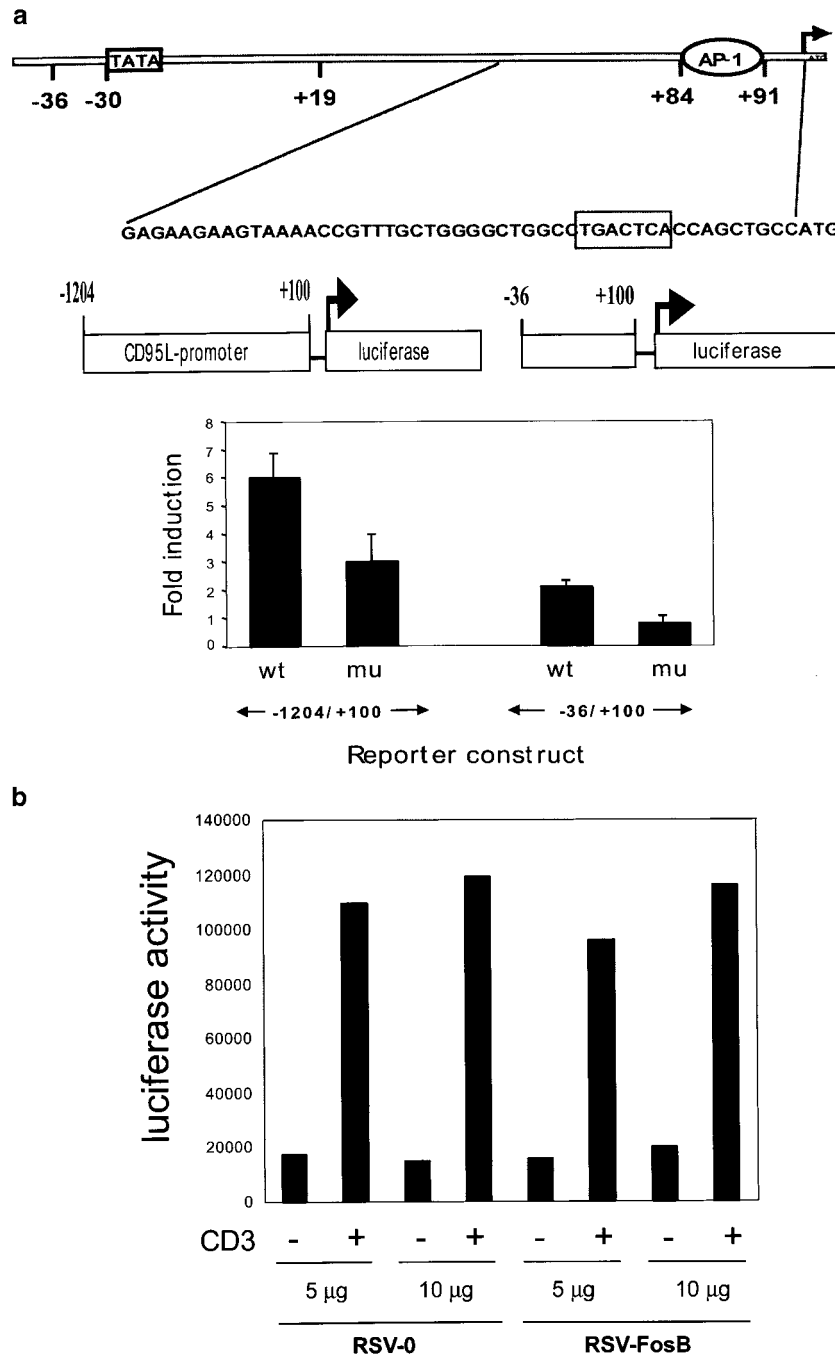


Figure 3 Mutated AP-1 strongly reduces TCR/CD3-mediated CD95L induction. **(a)** Jurkat T cells were transiently transfected with the $-1204/+100$ or the $-36/+100$ reporter construct containing either the wild-type (wt) or mutated (mu) AP-1 site in the 5' UTR (Eichhorst *et al.*, 2000) and were stimulated via TCR/CD3 for 12 h or left untreated. Luciferase activity was determined 48 h after transfection. The fold-induction levels given with respect to the untreated control culture are indicated. The results are representative of three independent experiments performed in triplicates **(b)** Jurkat T cells were transiently transfected with the $-1204/+100$ wild-type reporter construct in the presence of 5 μ g or 10 μ g of empty vector (c) or an expression plasmid encoding FosB and were stimulated via TCR/CD3 for 12 h or left untreated

binding site or expression of dominant negative mutants of c-Jun reduced TCR/CD3-mediated induction of CD95L expression and, in consequence, AICD.

Interestingly, in liver cells chemotherapeutic drugs also lead to activation of AP-1 and binding to the same site in the 5' UTR (Eichhorst *et al.*, 2000). DNA-damaging agents in human T lymphocytes (Kasibhatla

et al., 1998), during thymineless stress in TS(–) human colon carcinoma cells (Harwood *et al.*, 2000) and human liver cell lines, also result in the activation of AP-1 and subsequent CD95L expression. The AP-1-responsive site in T lymphocytes and colon carcinoma cells was identified upstream of the transcription site (Harwood *et al.*, 2000), whereas in liver cell lines the AP-1 site in

the 5' UTR was shown to be important (Eichhorst *et al.*, 2000). In all cases, a heterodimeric complex of c-Jun and c-Fos was suggested to mediate CD95L expression. Most interestingly, we demonstrate here that upon TCR/CD3 (re)stimulation, AP-1 complexes formed at the CD95L 5' UTR are composed of c-Jun and FosB but did not contain significant amounts of c-Fos. AP-1 binding was detectable on the sequence spanning -125 to -95 of the murine CD95L promoter upon TCR/CD3 stimulation, which depends on functional NF- κ B binding (Matsui *et al.*, 2000). The involvement of AP-1 in CD95L expression upon treatment with alkylating agents in fibroblasts derived from c-Jun-deficient mouse fetuses has also been described recently (Kolbus *et al.*, 2000). While c-Fos has been found to exhibit both pro- and antiapoptotic activities, depending on the cell type and stimulus, our data represent the first example of functional data on a critical involvement of FosB in CD95L-initiated apoptosis. Obviously, the type of stimuli dictates the abundance of Fos proteins showing very efficient induction of c-Fos in response to genotoxic agents (Kasibhatla *et al.*, 1998; Eichhorst *et al.*, 2000), whereas FosB appears to be more efficiently induced upon T-cell activation.

Most likely, FosB and c-Jun functionally synergize with other transcription factors, such as NF-AT, to facilitate regulated expression of CD95L. Mutation of the NF-AT binding site in the CD95L promoter completely blocks inducible transcription from this promoter (Li-Weber *et al.*, 1998; Rivera *et al.*, 1998). In cells of the immune system, cooperative NF-AT/AP-1 complexes are induced by stimulation of the antigen receptors of T and B cells (Rao *et al.*, 1997). Furthermore, AICD is also dependent on the cooperative activity of NF-AT/AP-1 (Macian *et al.*, 2000). Calcium mobilization activates the phosphatase calci-

neurin, which dephosphorylates NF-AT and causes its translocation from the cytoplasm to the nucleus (Crabtree, 1999). Since stimulation of the PKC/Ras pathway results in activation of AP-1 transcription complexes (Karin, 1995), cooperation between NF-AT and AP-1 proteins thus constitutes a powerful mechanism for signal integration of the calcium and PKC/Ras pathways. Cooperative binding has been demonstrated for the promoter region of several genes (Cockerill *et al.*, 1995; Rooney *et al.*, 1995). Possibly, NF-AT and AP-1 act together functionally by binding to different DNA sequences in the CD95L promoter. This might explain the weak induction of the -36/+100 reporter construct, which does not include the NF-AT recognition element (Li-Weber *et al.*, 1999). Regardless of the exact mechanism of cooperativity, our data have extended the number of AP-1 subunits that are involved in apoptosis. Whether or not other apoptotic processes, which are associated with changes in *fosB* expression, such as glutamate-mediated apoptosis of neuronal cells (Lidwell and Griffiths, 2000) and TGF β -induced apoptosis of hepatocytes and myeloid cells (Yamamura *et al.*, 2000), are also mediated through FosB/c-Jun-dependent activation remains to be determined.

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